A MODEL OSCILLATOR OF IRREGULAR STELLAR VARIABILITY

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Abstract. We have studied a set of equations with nonlinear and nonadiabatic terms which describes a simple oscillator. The equations have only one fixed point located at the origin. It is found that the oscillator shows the sequence of the period-doubling for the change of a parameter and results in chaotic oscillation. We illustrated the behaviour of the oscillator for several set of parameters and showed that the equations of the oscillator can be reduced to the one-zone model of stellar pulsation with simple nonlinear terms. It is suggested that the stellar irregular variability is resulted from the chaotic motion due to the nonlinear effect.

1. Introduction

Observationally nearly a half of pulsating stars in the Galaxy shows irregularity or semi-irregular behaviours (Hoffmeister *et al.*, 1985). Since these irregular stars have the characteristics of supergiants, it has been suggested that their extended atmospheres could reflect the acoustic wave regularly. If so, the irregularity will give us any more information but the bulkiness of the stars.

Recently, the nonlinear dynamics has been applied to this kind of stellar variability. Perdang (1985) reviewed the observation on the irregular variables and the theoretical studies. Baker *et al.* (1966) reported that the irregular pulsation is produced in the nonlinear one-zone model of stars. The chaotic behaviour in the one-zone model is also reported, when the model is dynamically unstable (Buchler and Regev, 1982; Auvergne and Baglin, 1985). Since the irregularity is observed in a lot of pulsating stars, it seems unlikely that only the stars in dynamically unstable state could show chaotic oscillations.

The nonlinear behaviour of adiabatic oscillators has been investigated as another candidate of irregular stellar pulsation (Perdang and Blacher, 1982, 1984; Aikawa and Whitney, 1987). The latter shows that an extremely high energy of oscillation is required for chaotic oscillations.

The chaotic behaviour in hydrodynamic simulations has been reported. Buchler and Kovacs (1987) recently indicate the sequence of period-doubling into chaotic oscillations in the hydrodynamic models of W Virginis stars. Takeuti (1987) pointed out that the numerical results of hydrodynamic models of less massive stars by several authors (Tuchman *et al.*, 1979; Fadeyev and Tutukov, 1981; Fadeyev, 1982, 1984; Nakata,

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1987) show a common feature that the amplitude of pulsations decreases suddenly after the slow increase of amplitude. It is reported that such a feature is different from the sequence of period doubling according to detailed studies in the first return map (Buchler *et al.*, 1987; Aikawa, 1988).

In the present paper, we study a set of equations of simple oscillator including nonlinear and nonadiabatic terms. In Section 2 we derive the nonlinear equations of the oscillator which is applicable to stellar atmospheric oscillation and discuss the stability of the equations. In Section 3 we illustrate the numerical solutions of the equations for several sets of parameters and show that the oscillator realizes the period doubling and finally becomes chaos. It is discussed in Section 4 that the oscillator is related to the Rossler equation and that the equation of the oscillator is an extended form with nonlinear terms of the one-zone stellar model. The conclusive remarks are given in the last section.

2. Equation of Oscillator with Nonlinear Nonadiabatic Terms

We consider a one-dimensional linear adiabatic oscillator of a layer in a star which is represented as

$$\mathrm{d}x/\mathrm{d}t = y\,, \tag{1}$$

$$\mathrm{d}y/\mathrm{d}t = \alpha x + z\,, \tag{2}$$

$$\mathrm{d}z/\mathrm{d}t = -\beta\,\mathrm{d}x/\mathrm{d}t\,, \tag{3}$$

where x and y are the displacement and the velocity, respectively, and z is the force acting on the oscillator which is defined by Equation (3). Equation (3) is the equation of state where β is proportional to the adiabatic exponent. We attempt to extend the equations to the nonadiabatic and nonlinear case.

In Equation (2), the equation of motion, we add a term μy for taking the effect of the γ-mechanism in the deep quasi-adiabatic region (see, for instance, Cox, 1980) or the influence of any other linear damping in the stellar envelope. Next, we add the nonadiabatic term F to the right-hand side of Equation (3) and assume that F is a function of only z and $\mathrm{d}z/\mathrm{d}t$. Considering the motion close to the adiabatic oscillation, we may replace $\mathrm{d}z/\mathrm{d}t$ by y. Then we may have

$$\mathrm{d}y/\mathrm{d}t = \alpha x + \mu y + z\,, \tag{4}$$

$$\mathrm{d}z/\mathrm{d}t = -\beta y - pz - qy + syz\,, \tag{5}$$

where p, q, and s are constant and the lowest nonlinear term is introduced. The terms $-pz - qy$ take a role of the phase shift of the cooling and heating to the change of z. The terms added in Equation (5) are responsible to the entropy change. In a case of s being negative, the nonlinear cooling (the outflow of radiation) occurs with a phase lag from the maximum contraction.

In order to analyze the characteristics of oscillator, the system of Equations (1), (4), and (5) for the oscillator is reduced into a third-order differential equation of x with the

nonlinear terms as

$$d^3x/dt^3 + A\, d^2x/dt^2 + B\, dx/dt + Cx = syz\,, \tag{6}$$

where $A = p - \mu$, $B = \beta - \alpha + q - \mu p$, and $C = -\alpha p$. Assuming the linearized equation of Equation (6) has the solution in the form of $\exp(\lambda t)$, then we have the cubic equation as

$$\lambda^3 + A\lambda^2 + B\lambda + C = 0\,. \tag{7}$$

When A and C is negligible compared with the unity and B, we have three real roots. In this case where the oscillation is adiabatic, the sign of B gives the dynamical stability of the system. Strictly, when the discriminant of the cubic equation is positive, the equation has three real roots. We, hereafter, restrict ourselves in the case where Equation (7) has one real root λ_1 and a pair of complex conjugate roots $a \pm ib$. As pointed out by Baker (1966), the condition of secular stability is $\alpha p < 0$ in which the real root λ_1 results to be negative. It is easily seen that the sign of the real part of the complex roots a depends on that of $AB - C$. The system is pulsationally unstable, when $AB - C < 0$ or

$$\beta + q < \mu p - \mu\alpha/(p - \mu)\,, \quad \text{for} \quad p - \mu > 0\,. \tag{8}$$

In the case of pulsational instability the origin is the unstable center of spiral, while the oscillator will leave the origin by the increase of excess heat loss or gain in the secular instability. More detailed characteristics of the fixed point of the equations is discussed by Takeuti *et al.* (1988).

It should be noted that the nonlinear term in Equation (6) can be represented by only x by using Equation (5). Then it is seen that the period of oscillation is modified by x and y and that y controls the damping as the resultants of modifications of cooling. We also note on the fixed points. By vanishing the time derivatives in the set of equations or Equation (6), we can see the fixed point is only one and locates at the origin in the (x, y, z) space except $p = 0$ and $\mu = 0$. For the case, integrating Equation (6) with nonlinear terms thus obtaining

$$d^2x/dt^2 + \{-(s/2)\, dx/dt\}\, dx/dt + \{B + (s\alpha/2)x\}x = 0\,, \tag{9}$$

we find that the system has two fixed points. We shall, however, concentrate to analyze the behaviour of the oscillator in the case of the only one fixed point.

3. Numerical Results

We numerically integrate the set of Equations (1), (4), and (5) in the vicinity of the origin for the case of dynamic and secular stability and pulsational instability. The orbits of the oscillator show the sequence of the period-doubling and chaotic behaviour for suitable combinations of parameters. We demonstrate the role of the parameters, that is, the effect of nonadiabaticity and nonlinearity on the development of chaos. First, we choose $\alpha = -0.5$, $\beta = \mu = q = 0.5$, and $s = 1$, as this set is convenient to compare the

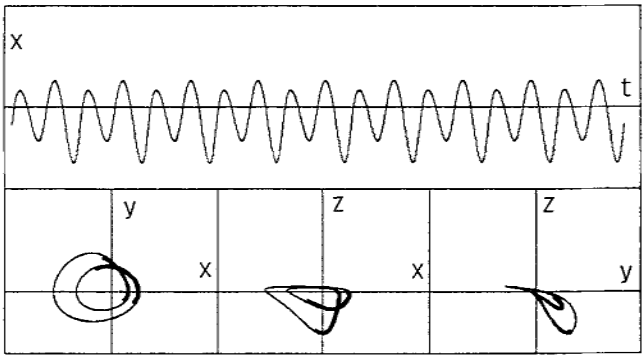

Fig. 1. The typical period 2 limit cycle for $\alpha = -0.5$, $\beta = 0.5$, $\mu = 0.5$, $p = 3.2$, $q = 0.5$.

present equations with the Rössler (1976) equation. As is seen from Equation (6), the amplitude of oscillations is restricted in a domain by the strong cooling when y becomes large. The boundary of the domain is determined by the value of p. The decrease of p, therefore, effectively corresponds to the increase of damping. In fact, the maximum values of y become small as p decreases.

In Figure 1, we present the orbit of the period 2 limit cycle where the difference between two orbits after the bifurcation is clearly illustrated. The thick line in Figure 1 indicates the portions where the nonadiabatic term $F(= -pz - qy + syz)$ is greater than zero, that is, the layer is heated, while the portion represented by the thin line shows the cooling in the layer. It is seen that the large amplitude orbit changes into the small amplitude one near around the region where $x > 0$, $y > 0$, and $z < 0$. The difference between the orbits occurs from the nonlinearity of F at the region where $x < 0$, $y > 0$, and $z < 0$. The nonlinearity affects every oscillations and can produce the chaotic one.

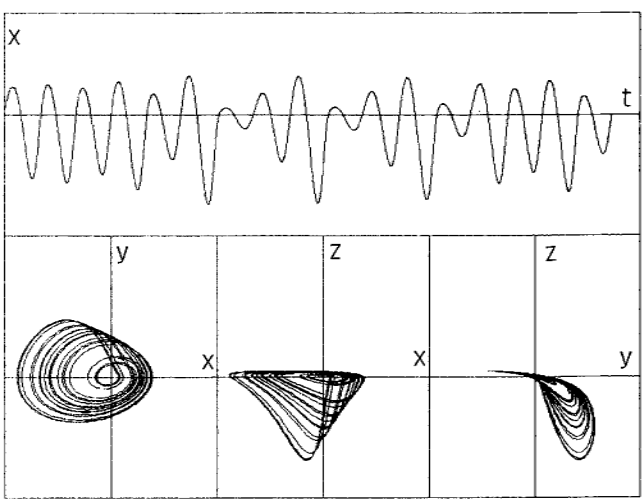

Fig. 2. A chaotic oscillation for $\alpha = -0.5$, $\beta = 0.5$, $\mu = 0.5$, $p = 4.0$, $q = 0.5$.

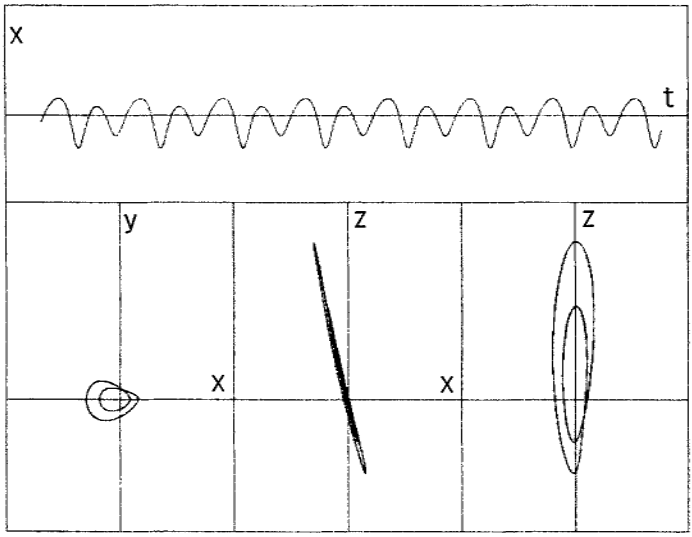

Fig. 3. The typical period 2 limit cycle for $\alpha = 4$, $\beta = 5$, $\mu = -0.5$, $p = -0.06$, $q = -0.7$, $s = 5$.

Trajectory of chaotic oscillation is illustrated in Figure 2, which is realized through a sequence of period-doubling.

The situation for the set of parameters with positive α is the same as before. We choose $\alpha = 4$, $\beta = 5$, $\mu = -0.5$, $q = -0.7$, $s = 5$, applying the oscillator to the one-zone stellar model. We also find the sequence of period-doubling for a series of p. In Figure 3 the typical period 2 limit cycle is shown. In Figure 4, we show the Feigenbaum tree by using the local maximum values of y, which clearly shows the bifurcation of orbits for the change of p.

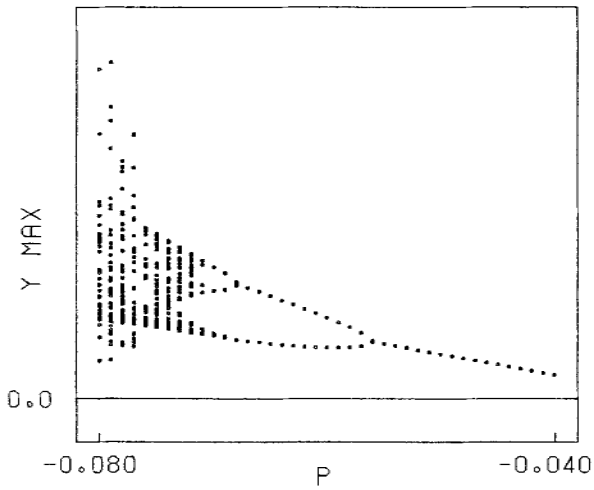

Fig. 4. The Feigenbaum tree for the same values of parameters as Figure 3 but p which is variable. The ordinate is the local maximum values of y.

We note that s is a scaling factor on which the amplitudes of orbits basically depend. When the (x, y, z) space is inverted, the system of the equations is unchanged if s is also replaced by $-s$.

Finally, we attempted to obtain the orbits of the oscillator in the case of dynamic and pulsational stability, but secular instability. Here it is noted only that the oscillatory expansion or contraction occurs and sometimes the final orbits of the limit cycle with period one centered off the origin is realized.

4. Discussion

4.1. Comparison with the Rossler equation

The Rössler equation is well-known as it is a simple prototype equation, while it shows various types of nonlinear orbits for the choice of parameters. The original Rossler equation is as

$$\mathrm{d}X/\mathrm{d}t = -Y - Z, \qquad \mathrm{d}Y/\mathrm{d}t = X + aY, \qquad \mathrm{d}Z/\mathrm{d}t = b + Z(X - c), \quad (10)$$

where a, b, and c are constants. Equation (9) can be arranged to the form in which two variables express the displacement and the velocity. Then moving one of two singularities to the origin, we have

$$\mathrm{d}x/\mathrm{d}t = y, \quad (11)$$

$$\mathrm{d}y/\mathrm{d}t = -x + ry + z, \quad (12)$$

$$\mathrm{d}z/\mathrm{d}t = (z - q)(-rx + y - pr) - pqr, \quad (13)$$

where p, q, and r are constants (Tanaka and Takeuti, 1987, unpublished).

The similarity is found between Equation (12) and Equation (4), but α is fixed to -1. It is interesting that the nonlinear behaviour is observed in both oscillators even though the third equation of the present oscillator, Equation (5) is simple in comparison with Equation (13).

4.2. Comparison with one-zone stellar models

The importance of one-zone stellar models for studying chaotic oscillation is first pointed out by Baker *et al.* (1966). They reported that various irregular pulsations were found in their one-zone model. The linear equations of one-zone stellar model for pulsation (Baker, 1966) can be written as

$$\mathrm{d}x/\mathrm{d}t = y, \quad (14)$$

$$\mathrm{d}y/\mathrm{d}t = 4x + z, \quad (15)$$

$$\mathrm{d}z/\mathrm{d}t = -3\gamma y - \varepsilon(\chi_x x + \chi_z z), \quad (16)$$

where χ_x and χ_z are the coefficients controlling the cooling and heating of the envelope determined by the characteristics of opacities, and γ and ε indicate the adiabatic

exponent and the degree of nonadiabaticity, respectively. It is obvious that the model is in dynamic and secular stability when $\gamma > \frac{4}{3}$ and $\chi_x > 4\chi_z$, and pulsationally stable for $\chi_x < 3\gamma\chi_z$.

Buchler and Regev (1982) and Auvergne and Baglin (1985) demonstrate the chaotic behaviour surrounding the singularity that is dynamically unstable, as they put $\gamma < \frac{4}{3}$ for the equilibrium state. Far from the equilibrium point their models have so large effective γ that oscillations become pulsationally stable. Different from their models, the present model gives the chaotic oscillation surrounding the dynamically stable equilibrium state.

We arrange the present system of equations into the form of one-zone models in order to see directly the fitness between them. Eliminating the term μy and supposing $\alpha = 4$ and $\beta = 3\gamma$, we have

$$\mathrm{d}x/\mathrm{d}t = y\,, \tag{17}$$

$$\mathrm{d}y/\mathrm{d}t = \alpha x + z\,, \tag{18}$$

$$\mathrm{s}z/\mathrm{d}t = -\beta y + m_x x + m_z z + (n_0 + n_1 z + n_2 y)y\,, \tag{19}$$

where $m_x = 4\mu$, $m_z = p - \mu$, $n_0 = \mu p - q$, $n_1 = s$, and $n_2 = -\mu s$. We can see how the damping coefficient μ inserted into the equation of motion plays the role for the entropy change. We illustrate a period 5 limit cycle as an example of chaos in Figure 5. The

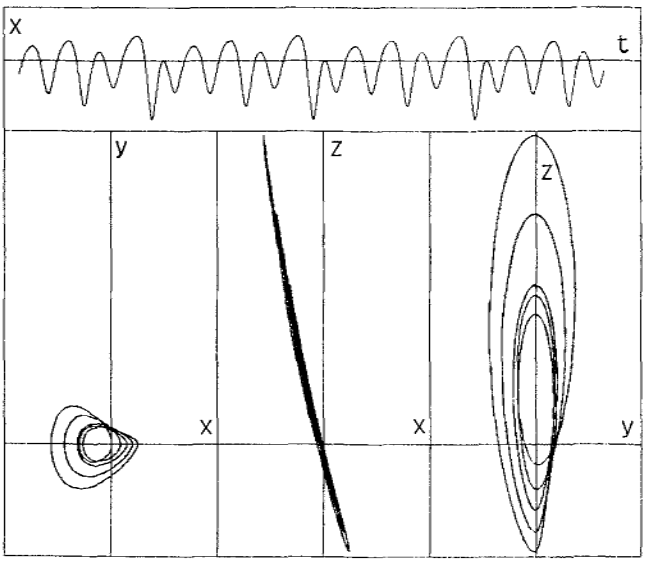

Fig. 5. The period 5 limit cycle as an illustration of chaos for the one-zone stellar pulsation model. Parameters: $m_x = 4\mu = -2$, $m_z = \mu - p = -0.4195$, $n_0 = \mu p - q = 0.725$, $n_1 = s = -5$, and $n_2 = -\mu s = -2.5$.

nonadiabatic terms on the right-hand side of Equation (19) consist of two groups: $m_x x + m_z z$ may express the effect of the κ-mechanism; $(n_0 + n_1 + n_2)y$ can do the nonlinear cooling with the phase lag from the maximum contraction when both n_1 and n_2 are negative.

4.3. Nonadiabatic terms for the chaotic motion

The combination of these coefficients determines where the entropy increases or decreases on the $y - z$ plane. The actual physical processes working in the models are not specified in the present study, but the characteristics required for producing the chaotic motion is indicated. That is, it is shown in numerical experiments that the increase of the pulsational instability makes oscillations to be chaotic.

The terms of a function of only x and z seem to be insufficient to produce the chaotic motion with lower order of nonlinearity. The effect of phase delay that is expressed by the term of y such as given in the present oscillator is necessary for simulating irregular stellar variability. The present oscillator is not the sole one producing chaos, but any others we have known may be more complicate.

5. Conclusion

We studied a simple oscillator which has rather physical definition and includes nonlinear nonadiabatic terms expressing the effect of phase delay in the entropy change. The oscillator shows the sequence of period-doubling into chaos in spite of its simple form. We could reduce the equations of the oscillator into a convenient form for comparing with those for pulsation of one-zone stellar models and, for suitable combination of parameters, we can expect irregular chaotic motion. The one-zone models including the opacities as the function of density and temperature in the second order seem to be insufficient to yield chaotic motion. It is suggested to study the effect of the phase delay resulted from the outermost layers of stars for simulating irregular pulsations.

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References

Aikawa, T.: 1988, *Astrophys. Space Sci.* **139**, 281.
Aikawa, T. and Whitney, Ch.: 1987, preprint.
Auvergne, M. and Baglin, A.: 1985, *Astron. Astrophys.* **142**, 388.
Baker, N. H.: 1966, in R. F. Stein and A. G. W. Cameron (eds.), *Stellar Evolution*, Plenum Press, New York, p. 333.
Baker, N. H., Moore, D. W., and Spiegel, E. A.: 1966, *Astron. J.* **71**, 844.
Buchler, J. R. and Kovacs, G.: 1987, *Astrophys. J.* **320**, L57.
Buchler, J. R. and Regev, O.: 1982, *Astrophys. J.* **263**, 312.
Buchler, J. R., Goupil, M.-J., and Kovacs, G.: 1987, *Phys. Letters* **126**, 127.
Cox, J. P.: 1980, *Theory of Stellar Pulsation*, Princeton University Press, Princeton, N.J.
Fadeyev, Yu.: 1982, *Astrophys. Space Sci.* **86**, 143.
Fadeyev, Yu.: 1984, *Astrophys. Space Sci.* **100**, 329.
Fadeyev, Yu. and Tutukov, V. A.: 1981, *Monthly Notices Roy. Astron. Soc.* **195**, 811.

Hofmeister, C., Richter, G., and Wenzel, W.: 1985, *Variable Stars,* translated into English by S. Dunlop, Springer-Verlag, Berlin.
Nakata, M.: 1987, *Astrophys. Space Sci.* **132**, 337.
Rössler, O. E.: 1976, *Phys. Letters* **57A**, 397.
Perdang, J.: 1985, in J. R. Buchler, J. M. Perdang, and E. A. Spiegel (eds.), *Chaos in Astrophysics*, D. Reidel Publ. Co., Dordrecht, Holland, p. 11.
Perdang, J. and Blacher, S.: 1982, *Astron. Astrophys.* **112**, 35.
Perdang, J. and Blacher, S.: 1984, *Astron. Astrophys.* **136**, 263.
Takeuti, M.: 1987, *Astrophys. Space Sci.* **136**, 129.
Takeuti, M., Tanaka, Y., and Okazaki, K.: 1988, *Sci. Reports Tohoku Univ.* (submitted).
Tanaka, Y. and Takeuti, M.: 1987, unpublished.
Tuchman, Y., Sach, N., and Barkatt, Z.: 1979, *Astrophys. J.* **234**, 217.